

Question 01

t	$v(t)$
2	10
4	20
6	25

① Degree, $n = 3 - 1 = 2$

$$P_2(t) = a_0 t^0 + a_1 t^1 + a_2 t^2$$

$$\begin{array}{lcl} 2^0 a_0 + 2^1 a_1 + 2^2 a_2 = 10 & \Rightarrow & 1a_0 + 2a_1 + 4a_2 = 10 \\ 4^0 a_0 + 4^1 a_1 + 4^2 a_2 = 20 & \Rightarrow & 1a_0 + 4a_1 + 16a_2 = 20 \\ 6^0 a_0 + 6^1 a_1 + 6^2 a_2 = 25 & \Rightarrow & 1a_0 + 6a_1 + 36a_2 = 25 \end{array}$$

$$\begin{vmatrix} 1 & 2 & 4 \\ 1 & 4 & 16 \\ 1 & 6 & 36 \end{vmatrix} \begin{vmatrix} a_0 \\ a_1 \\ a_2 \end{vmatrix} = \begin{vmatrix} 10 \\ 20 \\ 25 \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} a_0 \\ a_1 \\ a_2 \end{vmatrix} = \begin{vmatrix} -5 \\ 35/4 \\ -5/8 \end{vmatrix}$$

$$P_2(t) = -5 + \frac{35}{4}t - \frac{5}{8}t^2$$

$$P_1'(t) = \frac{35}{4} - \frac{5}{4}t$$

$$P_2''(t) = -\frac{5}{4} \quad \therefore \text{Acceleration, } P_2''(8) = -\frac{5}{4} = -1.25 \text{ ms}^{-2}$$

⑥

$$P_2(t) = l_0(t) f(t_0) + l_1(t) f(t_1) + l_2(t) f(t_2)$$

$$l_0(t) = \frac{(t-t_1)(t-t_2)}{(t_0-t_1)(t_0-t_2)} = \frac{(t-4)(t-6)}{(2-4)(2-6)} = \frac{(t-4)(t-6)}{8}$$

$$l_1(t) = \frac{(t-t_0)(t-t_2)}{(t_1-t_0)(t_1-t_2)} = \frac{(t-2)(t-6)}{(4-2)(4-6)} = \frac{(t-2)(t-6)}{-4}$$

$$l_2(t) = \frac{(t-t_0)(t-t_1)}{(t_2-t_0)(t_2-t_1)} = \frac{(t-2)(t-4)}{(6-2)(6-4)} = \frac{(t-2)(t-4)}{8}$$

$$P_2(t) = \frac{(t-4)(t-6)}{8} \times 10 + \frac{(t-2)(t-6)}{-4} \times 20 + \frac{(t-2)(t-4)}{8} \times 25$$

② If a new data point is added in the above scenario, we need to use Newton's divided difference method. Because we don't have to calculate everything again in this method, just calculate the things for newly added data point.

The degree of the new polynomial = $4-1=3$

②

Question 02

$$f(x) = x \sin x$$

(a)

	x	$f(x)$
x_0	$-\pi/2$	$-\frac{\pi}{2} \sin(-\frac{\pi}{2}) = \frac{\pi}{2}$
x_1	0	$0 \cdot \sin(0) = 0$
x_2	$\frac{\pi}{2}$	$\frac{\pi}{2} \sin(\frac{\pi}{2}) = \frac{\pi}{2}$

$$\text{Degree, } n = 3 - 1 = 2$$

$$P_2(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$$

$$= \frac{\pi}{2} - 1(x + \frac{\pi}{2}) + \frac{2}{\pi}(x + \frac{\pi}{2})x$$

$$= \frac{\pi}{2} - x - \frac{\pi}{2} + (\frac{2x}{\pi} + 1)x$$

$$= -x + \frac{2x^2}{\pi} + x$$

$$= \frac{2}{\pi}x^2$$

$$\begin{aligned}
 x_0 &= -\frac{\pi}{2} & f[x_0] &= \frac{\pi}{2} \\
 x_1 &= 0 & f[x_1] &= 0 \\
 x_2 &= \frac{\pi}{2} & f[x_2] &= \frac{\pi}{2} \\
 x_3 &= \pi & f[x_3] &= 0
 \end{aligned}$$

$$\begin{aligned}
 f[x_0, x_1] &= \frac{f[x_1] - f[x_0]}{x_1 - x_0} = \frac{0 - \pi/2}{0 + \pi/2} = -1 \\
 f[x_1, x_2] &= \frac{f[x_2] - f[x_1]}{x_2 - x_1} = \frac{\pi/2 - 0}{\pi/2 - 0} = 1 \\
 f[x_2, x_3] &= \frac{f[x_3] - f[x_2]}{x_3 - x_2} = \frac{0 - \pi/2}{\pi - \pi/2} = -1
 \end{aligned}$$

$$\begin{aligned}
 & \left. \begin{aligned} f[x_0, x_1] &= -1 \\ f[x_1, x_2] &= 1 \end{aligned} \right\} f[x_0, x_1, x_2] \\
 & \left. \begin{aligned} f[x_1, x_2] &= 1 \\ f[x_2, x_3] &= -1 \end{aligned} \right\} f[x_1, x_2, x_3]
 \end{aligned}$$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = \frac{1 + 1}{\pi/2 + \pi/2} = \frac{2}{\pi}$$

$$\left. \begin{aligned} f[x_0, x_1, x_2] &= \frac{2}{\pi} \\ f[x_1, x_2, x_3] &= \frac{-2}{\pi} \end{aligned} \right\} f[x_0, x_1, x_2, x_3]$$

$$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1} = \frac{-1 - 1}{\pi - 0} = \frac{-2}{\pi}$$

$$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0} = \frac{-\frac{2}{\pi} - \frac{2}{\pi}}{\pi + \frac{\pi}{2}} = -\frac{8}{3\pi^2}$$

$$\textcircled{b} \quad x = \frac{\pi}{3}$$

$$\begin{aligned} p_2\left(\frac{\pi}{3}\right) &= \frac{2}{\pi} \left(\frac{\pi}{3}\right)^2 \\ &= \frac{2}{\pi} \cdot \frac{\pi^2}{9} \\ &= \frac{2\pi}{9} \end{aligned}$$

$$\textcircled{c} \quad \text{new node} = \pi$$

$$f(\pi) = \pi \sin \pi = 0$$

$$n = 4 - 1 = 3$$

$$\begin{aligned} p_3(x) &= f[x_0] + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1) \\ &\quad + f[x_0, x_1, x_2, x_3](x-x_0)(x-x_1)(x-x_2) \end{aligned}$$

$$= \frac{2}{\pi} x^2 - \frac{8}{3\pi^2} \left(x + \frac{\pi}{2}\right)(x+0) \left(x - \frac{\pi}{2}\right)$$

$$= \frac{2}{\pi} x^2 - \frac{8x}{3\pi^2} \left(x^2 - \frac{\pi^2}{4}\right)$$

$$= \frac{2x^2}{\pi} - \frac{8x^3}{3\pi^2} + \frac{8x\pi^2}{4 \cdot 3\pi^2}$$

$$= \frac{2x^2}{\pi} - \frac{8x^3}{3\pi^2} + \frac{2x}{3}$$

Question 03

$$f(x) = e^{3x} - e^{-3x}$$

$$x \in [-1.5, 1.5]$$

$$x_0 = -1, x_1 = 0, x_2 = 1$$

$$f(x) = e^{3x} - e^{-3x}$$

$$f'(x) = 3e^{3x} + 3e^{-3x}$$

$$f''(x) = 9e^{3x} - 9e^{-3x}$$

$$\begin{aligned} f'''(x) &= 27e^{3x} + 27e^{-3x} \\ &= 27(e^{3x} + e^{-3x}) \end{aligned}$$

$$\begin{aligned} f'''(1.5) &= 27(e^{3 \times 1.5} + e^{-3 \times (-1.5)}) \\ &= 4860.92509 \end{aligned}$$

$$w(x) = (x - x_0)(x - x_1)(x - x_2)$$

$$= (x + 1)(x - 0)(x - 1)$$

$$= x(x^2 - 1)$$

$$= x^3 - x$$

$$w'(x) = 3x^2 - 1; x_1 = \frac{1}{\sqrt{3}}, x_2 = -\frac{1}{\sqrt{3}}$$

x	$w(x)$
$\frac{1}{\sqrt{3}}$	-0.38490
$-\frac{1}{\sqrt{3}}$	0.38490
-1.5	-1.875
1.5	1.875

$$\text{error bound} = \left| \frac{f^3\left(\frac{2}{3}\right)}{3!} \right| |(x-x_0)(x-x_1)(x-x_2)|$$

$$= \left| \frac{4860.92509}{3!} \right| \times |1.875|$$

$$= 810.1541817 \times 1.875$$

$$= 1519.039091$$

$$= 1519.0$$